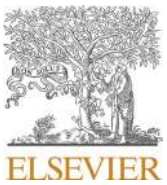
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An integrated deep learning approach for assessing the visual qualities of built environments utilizing street view images

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ABSTRACT

Investigating residents' visual preferences and perception of built environments is crucial in visual landscape assessment (VLA). While traditional methods face challenges in large-scale applications, the advancement of deep learning techniques and the availability of street view images (SVIs) present new opportunities. However, existing approaches for assessing SVIs' visual qualities are of lower precision, and the link between objective visual elements and subjective perceptions of SVIs remains unclear. In this study, we propose a novel deep learning approach, "SegFormer-B5 + ConvNeXt-B + RF", which achieves an average accuracy of 78.47% in predicting six subjective perceptions (beautiful, boring, depressing, lively, safe, and wealthy) within the Place Pulse 2.0 dataset. This provides an effective tool for assessing citizens' visual perceptions of urban environments. Subsequently, to demonstrate its practical application, we conducted a case study using 36,620 SVIs from the Tianhe District of Guangzhou. Perception maps were constructed based on four objective metrics and six subjective metrics. Results showed a correlation between the spatial distribution of objective visual elements and subjective perceptions, with city centers generally perceived more positively than suburbs. Our application of SHapley Additive exPlanation (SHAP) and Class Activation Map (CAM) visualizations yielded interpretable insights consistent with eye-tracking studies, highlighting human focus on artificial objects, attractive and unattractive elements, and heterogeneous landscapes. It's noteworthy that urban planners and decision-makers in other cities can apply our approach to generate perception maps that identify low-quality areas. SHAP and CAM visualizations further assist in understanding which aspects draw human attention in these areas. This knowledge is crucial for urban designers to implement targeted renewal strategies, ultimately contributing to the creation of sustainable and living-friendly cities.

1. Introduction

The European Landscape Convention defines a landscape as an area perceived by locals or visitors, whose visual features and character emerge from the action of natural and/or cultural (that is, human) factors (Council of Europe, 2000). As an entity perceived by people, the landscape is no longer a purely organized, managed, or even administrative territory. It gains meaning through individual and societal interpretations and therefore has perceptive, aesthetic, artistic, and existential significance (Antrop, 2005; Butler and Berglund, 2014). Visual landscape assessment (VLA), which focuses on the mechanisms of interaction between landscape and humans, refers to the value judgments made by humans using visual senses on the external and functional aspects of the landscape (Daniel, 2001). The field is interested in exploring the interaction between objective visual elements and residents' subjective perceptions (Kaplan and Kaplan, 1989). Objective elements include all tangible components, both natural and man-made, which form the foundation for human perceptions. The measurement of subjective perceptions links directly to objective elements, aiding in defining and validating the meanings of visual elements in our environment (Wendel et al., 2012).

Related research has developed various theories and methods since the 1960s, such as the four paradigms (expert, psychophysical, cognitive, and experiential) proposed by Zube et al. (1982), the four

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landscape perception metrics (coherence, mystery, complexity, and legibility) established by Kaplan (1975), and the Scenic Beauty Estimation method proposed by Daniel (1976). These studies have revealed how human perception works, helping to address broader issues of human well-being, urban design, and environmental action (Gobster et al., 2019), pivotal for creating living-friendly and sustainable urban spaces. However, traditional methods like questionnaires and in-depth interviews for photo-mediated assessments (Scott and Benson, 2002) are labor-intensive, costly, time-consuming, and limited in data sources, posing challenges for large-scale applications in built environments.

In the era of big data, the proliferation of street view images (SVIs) from platforms like Google and Baidu Map, combined with advancements in deep learning and enhancements in computer capabilities have made SVIs a vital resource. They accurately capture the spatial characteristics of geographic spaces, proving invaluable in assessing the visual quality of large-scale built environments (Li et al. 2017, Zhang, Zhang et al., 2018, Kang et al., 2020, Biljecki and Ito, 2021). Recent review studies (Ibrahim et al., 2020, Kang et al., 2020, Biljecki and Ito, 2021; He and Li, 2021) have also highlighted the growing importance of SVIs in perception assessments, a core aspect of VLA studies. This underscores the evolving significance of SVIs-based visual assessment in contemporary research.

In VLA research based on SVIs, semantic segmentation models are commonly employed to extract objective visual elements, while image classification models are utilized for predicting subjective perceptions. Nevertheless, the accuracy of widely-used models in this domain does not align with the advancements in mainstream deep learning research. For example, widely-used semantic segmentation models like FCN (Dai et al., 2021, Ki and Lee, 2021), DeepLabV3+ (Ji et al. 2021, Xia et al., 2021), PSPNet (Zhang et al., 2018), and Segnet (Tang and Long, 2019, Zhou et al., 2019) falls short when compared to the latest SegFormer-B5 model (Appendix A), whose capability of extracting objective elements from SVIs is not well-explored, and the comparative recognition capabilities of these models remain under-researched. This leads to our first question: how much better can we achieve in recognizing objective visual elements within SVIs?

Furthermore, for ImageNet-1K image classification, contemporary models like ConvNeXt (Liu et al., 2022), Swin, and ViT show top-1 accuracies of over 85% (Appendix B), outperforming earlier models. But the application of these advanced models in perception prediction within the VLA framework is limited. This raises our second question: can the incorporation of the advanced models facilitate a better evaluation of subjective perceptions within SVIs?

Finally, existing studies often focus solely on either extracting objective visual elements or subjective perceptions. Several studies (Ji et al. 2021, Chen et al., 2022) have employed regression analysis to probe the correlation between these two aspects. However, such analyses are insufficient and less comprehensive. So our third question emerges: besides correlation analysis, what are the more effective and interpretable ways to validate the relationship between the two? And which aspects of SVIs are of particular interest to humans?

In this paper, we aim to introduce a precise deep-learning approach for assessing visual qualities in large-scale built environments, integrating both objective visual elements and subjective perceptions. We conduct extensive comparative and ablation studies to demonstrate the effectiveness of our proposed methods. To exemplify the practical application of this approach, we have chosen Guangzhou city as a case study area. This implementation focuses on examining the distribution patterns of key metrics related to visual qualities, aiming to pinpoint areas that require urgent renovation. Additionally, we employ SHAP (SHapley Additive exPlanations) and CAM (Class Activation Mapping) visualizations to gain deeper insights into the elements that attract human attention in these low-quality areas. These valuable insights could serve as a foundation for implementing highly effective strategies aimed at promoting the development of high-quality and sustainable cities.

2. Related work

Given the complex interplay of factors within the visual environment, researchers have developed various evaluation frameworks to assess the impact of multiple indicators on visual quality. As summarized by Daniel (2001), traditional VLA research typically falls into two paradigms: objective and subjective. The objective paradigm quantifies visual quality through the physical characteristics of the environment as measured by experts, whereas the subjective paradigm evaluates quality based on the observer's visual perception.

Contemporary VLA research endeavors to synthesize these paradigms to provide a more holistic view of visual quality. While these studies are based on many principles from earlier methodologies (Gobster et al., 2019), they also incorporate more advanced and precise techniques. For instance, eye-tracking technology (Sussman and Ward, 2019) is employed to track human visual attention, while large-scale online surveys (Sahraoui et al., 2021) are more combined with natural language processing (Liu et al., 2021) to gain a deeper understanding of spatial preferences and visual perceptions. Especially remarkable is the applications of big data from SVIs coupled with artificial intelligence, which addresses the limitations of traditional methodologies like high costs and restricted geographic coverage. It enables rapid, effective, and objective evaluations of built environments from the human-eye perspective. Central to these applications are the deep learning tasks of semantic segmentation and image classification, which are crucial in enabling a thorough and intricate analysis of the built environments.

2.1. Extracting objective visual elements using semantic segmentation

Objective visual elements extracted from urban appearances form the basis of residents' visual and psychological impressions of the city (Naik et al., 2016). Semantic segmentation, which classifies each pixel into a distinct category (Guo et al., 2019), offers greater accuracy in identifying street elements compared to traditional methods (Kang et al., 2020). Models trained on datasets like Cityscapes can automatically identify elements such as plants, buildings, and sky in SVIs. By analyzing the proportions of these elements, metrics like sky view factor (Xia et al., 2021), green view index (Helbich et al. 2019, Ki and Lee, 2021), walkability (Zhou et al., 2019, Koo et al., 2022), and enclosure (Yin and Wang, 2016, Wang et al., 2019) can be calculated. These metrics enable a precise and efficient quantitative description of the detailed composition of the physical environment.

2.2. Predicting subjective perception using image classification

Subjective perceptions of SVIs are often predicted using the MIT Place Pulse 2.0 dataset (Dubey et al. 2016). Researchers typically develop image classification models (Porzi et al. 2015) utilizing this dataset, which consists of two types of deep learning tasks. The first is a binary classification task, such as dividing image scores into "positive" and "negative" binary labels, and then using models like ResNet or SVM to predict perception scores (Zhang et al., 2018, Chen et al., 2022, Wei et al., 2022). Although by doing these researchers can acquire perception scores, it is difficult to capture fine-grained details by directly dividing all scores into two major categories. Therefore, Ji et al. (2021) discrete the data and labels into ten classes and built a more accurate approach by using InceptionV3 and the random forest algorithm (Wendel et al., 2012). Table 1 summarizes the most relevant state-of-the-art methods concerning urban perception.

3. Methodology

3.1. Research framework

The research framework (Fig. 1) is structured into two main sections: model training and model application.

Table 1
Related work on measuring urban perception.

Work	Classification	Approach
Zhang et al. (2018)	binary	DCNN (ResNet) + SVM
Yao et al. (2019)	binary	FCN + RF
Wang et al. (2021)	binary	CNN (VGGNet)
Ji et al. (2021)	ten classes	DCNN (InceptionV3) + RF
Chen et al. (2022)	binary	SVM
Wei et al. (2022)	binary	ResNet
Xu et al. (2023)	binary	CNN (VGGNet)

Model Training Section: Train the proposed "SegFormer-B5 + ConvNeXt-B + RF" approach using the Place Pulse 2.0 dataset and evaluate its performance quantitatively through metrics like Mean Absolute Error (MAE) and accuracy. Subsequently, compare the model with recent state-of-the-art methods to demonstrate the effectiveness of our proposed approach.

Model Application Section: Using SVIs of Tianhe district in Guangzhou City as a case study, we apply the proposed "SegFormer-B5 + ConvNeXt-B + RF" approach to demonstrate its practical utility. We compare the model's predictions with the actual environmental conditions and investigate the links between objective visual elements and subjective perception scores. Our aim is to provide insights that could be valuable for urban construction and development.

3.2. Model training

3.2.1. Dataset

1) Place Pulse 2.0 Dataset (Dubey et al., 2016): Containing 110,988 SVIs collected from 2007 to 2012, and 81,630 online volunteers' perception scores on six dimensions of beautiful, boring, depressing, lively, safe, and wealthy. Participants need to compare two images at a time, choosing the one they prefer. We employ the Microsoft TrueSkill algorithm (Herbrich et al., 2006) to transform these pairwise comparisons into each image's perception score (focusing on images compared at least five times (Wei et al., 2022)). These scores are normalized from 0 to 10 using maximum-minimum normalization (Eq. (1)), and then binned into ten classes, from 0 to 9 (Eq. (2)).

$$\widehat{x}_{ik} = \left(\frac{x_{ik} - \min(x_i)}{\max(x_i) - \min(x_i)} \right) * 10 \tag{1}$$

$$C_{ik} = \begin{cases} class\ 0, 0 \le \widehat{x}_{ik} < 1 \\ class\ 1, 1 \le \widehat{x}_{ik} < 2 \\ \quad \dots \\ class\ 9, 9 \le \widehat{x}_{ik} < 10 \end{cases} \tag{2}$$

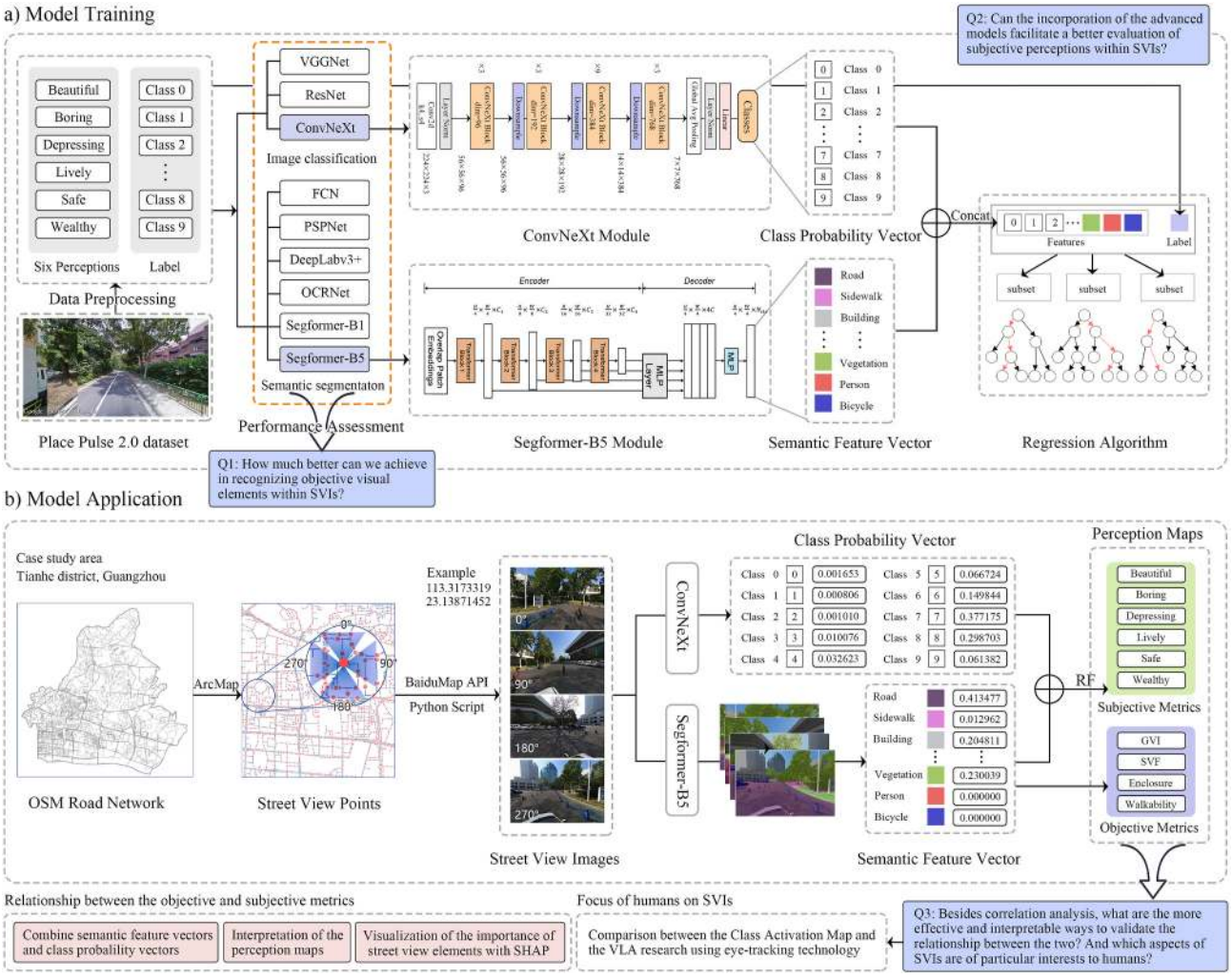


Fig. 1. a) Illustration of the training process for our approach using the Place Pulse 2.0 dataset. b) Application of our approach on the Tianhe District dataset, showcasing its practical utility.

where i is the ith perception of the six categories in the Place Pulse 2.0 dataset, x_{ik} is the ith image's perception score. $\hat{x}_{ik}$ and C_{ik} are the normalized scores and binned class of the kth image in the ith perception, respectively.

2) Cityscapes Dataset (Cordts et al. 2016): Containing 5000 high-resolution images and 5000 pixel-level object annotations from 50 cities, covering 19 categories such as roads, buildings, pedestrians, and vehicles (Appendix C).

3.2.2. Semantic segmentation

We use a SegFormer-B5 (Xie et al. 2021) based semantic segmentation model pre-trained on the Cityscapes dataset to perform pixel-level semantic segmentation. This process extracts the pixel-wise ratios $L = \{l_1, l_2, ..., l_n\}, l \in \mathbb{R}^{19}$ for each of the 19 object categories present in the image i_n.

$$s_n^0 = \mathrm{CNN}(i_n), \tag{3}$$

$$s_n^l = f_{seg}\left(\mathrm{MSA}\left(\mathrm{LN}\left(s_n^{l-1}\right)\right)\right), l = 1, 2, ..., L \tag{4}$$

$$l_n = \left[\frac{0 \text{ in } f_{seg}\left(s_n^l\right)}{\text{total pixels}}, \frac{1 \text{ in } f_{seg}\left(s_n^l\right)}{\text{total pixels}}, ..., \frac{18 \text{ in } f_{seg}\left(s_n^l\right)}{\text{total pixels}}\right] \tag{5}$$

where f_{seg} is a multi-layer perceptron (MLP) with 19 classes, and each l_n is a pixel-wise label map for i_n. CNN stands for Convolutional Neural Network. Each label map l_n contains pixel-wise ratios for each of the object categories in the SVIs.

3.2.3. Image classification

We employ the ConvNeXt-B (Liu et al., 2022) model to train classification models, aiming to compute a 10-dimensional vector $C = \{c_1, c_2, ..., c_n\}, c \in \mathbb{R}^{10}$ representing subjective perception class probabilities for each image i_n. This vector reflects the distribution of people's perception scores across ten specific SVI categories.

$$Y = \mathrm{LN}(\mathrm{GELU}(\mathrm{LN}(\mathrm{Conv}(i_n))) + i_n) \tag{6}$$

where LN refers to layer normalization, GELU stands for Gaussian Error Linear Unit, serving as an optimized activation function, Conv denotes a depth-wise separable convolution layer, which enhances the model's efficiency in processing image data.

3.2.4. Regression module

The subjective classification vector c_n and the objective feature-level semantic vector l_n are then concatenated and provided as multiscale input to a Random Forest regression model.

$$p_n = \text{Random Forest}([c_n : l_n]) \tag{7}$$

where p_n is a concrete number ranging from 0 to 10, this number represents the predicted human perceptual score for the given image. The Random Forest model has learned to map the concatenated feature vector to these scores based on the training data, effectively combining the image classification outcomes and the fine-grained semantic features.

During the training phase, we train perception models independently for each human perception index. However, during application, we ensemble these models for practical utility.

3.3. Model application

3.3.1. Tianhe District dataset

We select the Tianhe district of Guangzhou city as our case study area (Fig. 2b) to perform a real-world application of our approach in a specific urban environment. This district covers approximately 96.33 square kilometers, with 21 streets and a dense population of about 2.24 million people. The region is characterized by a distinct topography that descends from north to south, featuring low mountains in the north (222–400 m), a central plateau (30–50 m), and a low-lying southern plain (1.5–2 m). This varied topography leads to a noticeably diverse distribution of natural resources, resulting in significant variations in street environments. Moreover, a notable socioeconomic disparity exists. The southwest region is the core commercial area of Guangzhou. The further away from this area, the lower the economic level becomes. These variations offer a comprehensive context for examining metrics like "wealthy" and "lively" in our model. Besides, Guangzhou's subtropical climate, which ensures greenery throughout the year, helps reduce seasonal fluctuations in the SVIs (He and Li, 2021).

We first obtained the road network through OpenStreetMap and then generated street view points every 50 m using ArcMap 10.6, totaling 22,049 points (Fig. 2c). We then crawled the latest SVIs for these points from Baidu Map's static map API (https://lbs.baidu.com), which offers higher frequency, range, and resolution of SVIs in China compared to Google Map (Du et al. 2020, Biljecki and Ito, 2021). For comprehensive coverage, we collected four images at each point (0°, 90°, 180°, and 270° directions), each with a resolution of 480 × 320 pixels. This process yielded 36,260 effective SVIs.

3.3.2. Perception maps generation and analysis

We constructed perception maps by integrating six subjective metrics from the Place Pulse 2.0 dataset with four objective metrics. The proportion of each type of visual element extracted by semantic segmentation as P_i (Eq. (8)), and we calculated four widely-used objective visual metrics (Wang et al., 2019, Zhou et al., 2019, He and Li, 2021; Ki and Lee, 2021, Xia et al., 2021): Green View index (GVI), Sky View Factor (SVF), Enclosure and Walkability (Table 2).

$$P_i = \frac{C_i}{C_{sum}}, i \in \{\text{Road, Sidewalk, ..., Bicycle}\} \tag{8}$$

where i is one of the 19 element categories obtained through semantic segmentation, and P_i is the ratio of the pixel count of element i to the total pixel count C_{sum} in an image.

To effectively highlight high and low-quality areas based on perception maps and reveal spatial interconnections between subjective and objective metrics, we conducted global and local Moran's I analysis using ArcMap 10.6. The former metric ranges from 0 to 1, with values closer to 1 indicating stronger global spatial autocorrelation, which is used to describe the overall distribution status of a phenomenon. The latter metric reflects the spatial aggregation of local region variables, dividing the points into four categories: high-high (H–H) points, with high values both in themselves and their surroundings; low-high (L-H) points, with low values but high values in their surroundings; low-low (L-L) points, with low values both in themselves and their surroundings; and high-low (H-L) points, with high values but low values in their surroundings.

3.3.3. Visualization of street view elements' importance with SHapley additive exPlanations (SHAP)

We combine the semantic feature vectors generated by SegFormer-B5 with the class probability vectors generated by ConvNeXt-B to predict the perception scores. The relationship between the two can be visualized using the SHAP method (Lundberg et al., 2020). SHAP's core principle involves creating subsets of the training dataset through sampling. For each subset, the model's prediction changes are attributed to individual features. By averaging the Shapley values across all subsets, we ascertain the impact of each feature on the model's predictions. This method enables a visual exploration of how different street view elements contribute to perception scores.

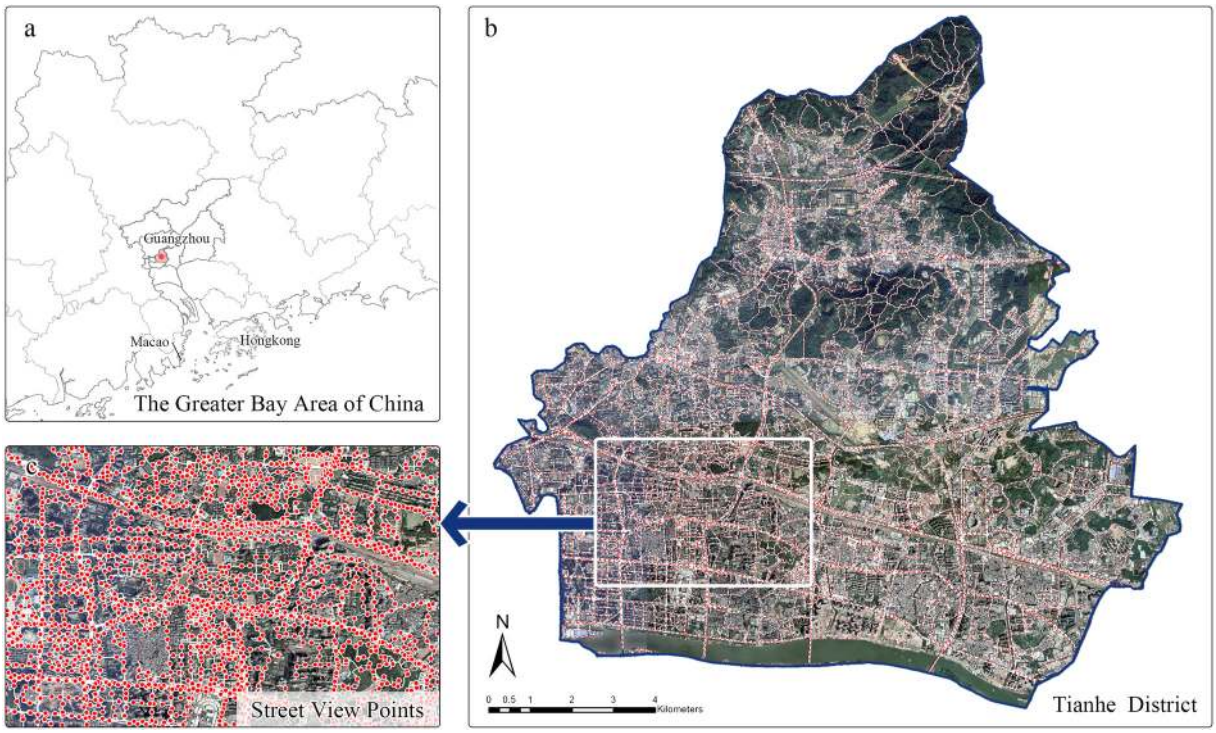


Fig. 2. Case study area. (a) The Greater Bay Area with emphasized locations of Guangzhou. (b) Tianhe District in Guangzhou, showcasing its road network. (c) A detailed segment of Tianhe District, highlighting points along the roads.

Table 2
Objective visual metrics.

Metrics	Formula	Introduction
Green View Index (GVI)	$GVI_i = P_{Vegetation}$	GVI (Yang et al., 2009) measures the visibility of vegetation as observed by the human eye, assessing urban streets' greenery levels (Ki and Lee, 2021). $P_{Vegetation}$ represents the vegetation ratio extracted from SVIs.
Sky View Factor (SVF)	$SVF_i = P_{Sky}$	SVF evaluates the openness of streets, which represents the visibility of the sky in human-eye observation (Li et al. 2018, Xia et al., 2021). P_{Sky} represents the sky ratio extracted from SVIs.
Enclosure	$Enclosure_i = P_{Building} + P_{Wall} + P_{Fence} + P_{Pole} + P_{Light} + P_{Sign} + P_{Vegetation}$	Enclosure quantifies how much the street environment surrounds pedestrians, which has a certain impact on their perceptions (Owens, 1993). It's calculated using vertical elements like buildings, walls, fences, and vegetation.
Walkability	$Walkability_i = \frac{P_{Fence} + P_{Sidewalk}}{P_{Road}}$	Defined by Ma et al. (2021) as the environmental visual impact from a horizontal perspective, influenced by the ratio of pedestrian-related traffic facilities (e.g., sidewalks, fences) to roads.

3.3.4. Visualization of Class Activation Map (CAM)

Since image classification models simulate the perception of SVIs by humans, we believe that it is particularly important to understand the focus of these models on SVIs. CAM can visualize the parts of an image that a deep learning model focuses on, showing us what it 'focuses on'. For ConvNetXt, the final convolutional layer's feature map can capture high-level semantic information of the input image. After processing with a global average pooling layer, we can obtain the weights of each feature map channel. These weights are visualized using the Grad-CAM method, enhancing the interpretability of the model and providing insights into which aspects of the SVIs are deemed significant by the model.

4. Experiments and results

4.1. Model training and evaluation

4.1.1. Training details

Subsequent to the execution of data cleaning and balancing procedures on the Place Pulse 2.0 dataset, the dataset was divided at a 7:2:1 ratio (Table 3). Image augmentation used RandomResizedCrop, RandomErasing, and RandomFlip. Optimization was via AdamW with Cosine Annealing schedular for learning rate adjustment, and Cross Entropy loss for loss computation.

4.1.2. Evaluation metrics

The effectiveness of these model combinations is evaluated using accuracy (ACC), mean squared error (MSE), mean absolute error (MAE), and R^2 (Eq. (9) to (12)). Lower MSE and MAE values signify higher accuracy, while a higher R^2 value indicates greater explanatory power of the model.

Table 3
Distribution of model training data.

	BeP.	BoP.	DeP.	LiP.	SaP.	WeP.
Training	7648	5652	6022	8039	7691	5928
Validation	2185	1614	1719	2296	2197	1694
Testing	1095	819	866	1150	1101	858
Total (No duplication):45,663						

BeP., BoP., DeP., LiP., SaP., WeP. refer to the perceptual dimensions of Beautiful, Boring, Depressing, Lively, Safe, and Wealthy, respectively.

$$ACC = \frac{N_{\hat{y}_i = y_i}}{n} \tag{9}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{10}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{11}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y_i})^2} \tag{12}$$

Where n is the number of SVIs, y_i is the actual perception score of the ith image. $\overline{y_i}$ is the average actual perception score. $\hat{y}_i$ is the prediction score. $N_{\hat{y}_i = y_i}$ is the number of images predicted correctly.

4.2. Comparative study of perception prediction models

Research on predicting subjective human perception has increasingly employed convolutional neural networks like ResNet (Zhang et al., 2018, Wei et al., 2022), InceptionV3 (Ji et al. 2021), and VGGNet (Wang et al., 2021, Xu et al. 2023). In our comparative study, we compared models like ResNet (He et al. 2016), VGGNet (Simonyan and Zisserman, 2014), and the latest ConvNeXt (Liu et al., 2022) models. Here, we fine-tuned ResNet50, VGG19 and ConvNext-B, initially pre-trained on the ImageNet-1K dataset (resolution = 384x384), on the Place Pulse 2.0 dataset to report the different performances across models.

As shown in Table 4, the "SegFormer-B5 + ConvNeXt-B + RF" approach we proposed proved the most effective, achieving an average ACC of 78.56%. It demonstrated substantial improvements over ConvNeXt alone, with MAE below 0.59829, MSE below 1.38704, and R^2 above 0.63794 for each perception. This suggests that the integration of objective visual elements notably enhances the prediction accuracy of subjective perceptions and indirectly underscores the connection between these two aspects. Besides, when using image classification models independently, ConvNeXt outperformed others, achieving an average ACC of 75.88%, surpassing VGGNet and ResNet by 1.63% and 5.06%, respectively.

To confirm the superiority of the Random Forest (RF) model in regression analysis, we merged semantic feature vectors from SegFormer-B5 with class probability vectors from ConvNeXt-B and evaluated various regression models, including Linear Regression (LR), Support Vector Regression (SVR), K-Nearest Neighbors Regression (KNR), AdaBoost, Gradient Boosting Machine (GBM), Bagging, Multi-Layer Perceptron (MLP) and RF. The RF model exhibited the most robust performance, as shown in Table 5, achieving the best results in all categories except for the MAE in DeP and LiP, the MSE in DeP and SaP, and the R^2 value for DeP. This underscores its overall superiority in the majority of the regression methods.

4.3. Comparative study of semantic segmentation models

Comparative studies examining the effectiveness of various models in recognizing the objective visual elements of SVIs are notably scarce. To address this gap, we selected a range of commonly used models like FCN (Dai et al., 2021, Ki and Lee, 2021), PSPNet (Zhang et al., 2018, Chen et al., 2022), and DeepLabV3+ (Ji et al. 2021, Xia et al., 2021), alongside the more recent models such as OCRNet and SegFormer (B1 and B5). All models were trained on the Cityscapes dataset. We then applied these models to perform semantic segmentation on SVIs from two different sources: 45,663 images from the Place Pulse 2.0 dataset and 36,620 images from Tianhe District datasets.

In Fig. 3, we visualized the segmentation results and observed that SegFormer-B5 outperformed other models on both the Place Pulse 2.0 and Tianhe District datasets. In the Tianhe District dataset, the differences among model performances were minimal. However, significant discrepancies were observed in the Place Pulse 2.0 dataset. We hypothesize that this variation is attributed to the models being trained on the Cityscapes dataset, which features high-resolution images (1024×2048 pixels), in contrast to the lower resolution (300×400 pixels) of the Place Pulse 2.0 images.

As Fig. 4 indicates, a consistent pattern emerged in both datasets regarding the average percentage composition of elements like roads, vegetation, sky, and buildings. To assess model differences, we

Table 4
Performance of each model combination in predicting perception scores.

	Model	BeP.	BoP.	DeP.	LiP.	SaP.	WeP.
ACC	SegFormer-B5 + RF	0.399	0.331	0.429	0.428	0.428	0.377
	VGG19	0.749	0.742	0.745	0.725	0.752	0.742
	ResNet50	0.709	0.688	0.712	0.714	0.713	0.712
	ConvNeXt-B	0.764	0.747	0.759	0.762	0.764	0.759
	SegFormer-B5 + ConvNeXt-B + RF (ours)	**0.770**	**0.786**	**0.786**	**0.774**	**0.791**	**0.807**
	Improvement	**0.021**	**0.044**	**0.041**	**0.048**	**0.040**	**0.065**
MAE	SegFormer-B5 + RF	1.345	1.607	1.257	1.333	1.260	1.351
	VGG19	0.704	0.665	0.576	0.624	0.642	0.661
	ResNet50	0.817	0.731	0.662	0.805	0.772	0.742
	ConvNeXt-B	0.652	0.598	0.544	0.662	0.616	0.605
	SegFormer-B5 + ConvNeXt-B + RF (ours)	**0.584**	**0.550**	**0.533**	**0.598**	**0.582**	**0.532**
	Improvement	**−0.120**	**−0.115**	**−0.042**	**−0.026**	**−0.060**	**−0.129**
MSE	SegFormer-B5 + RF	2.601	3.644	2.274	2.619	2.276	2.575
	VGG19	2.730	2.286	1.740	2.782	2.438	2.328
	ResNet50	3.151	2.275	1.993	3.261	2.936	2.576
	ConvNeXt-B	2.489	1.887	1.591	2.610	2.257	2.062
	SegFormer-B5 + ConvNeXt-B + RF (ours)	**1.387**	**1.210**	**1.076**	**1.427**	**1.297**	**1.163**
	Improvement	**−1.343**	**−1.076**	**−0.665**	**−1.354**	**−1.141**	**−1.165**
R2	SegFormer-B5 + RF	0.002	−0.007	0.006	0.015	0.013	0.008
	VGG19	0.390	0.386	0.412	0.367	0.472	0.381
	ResNet50	0.296	0.389	0.327	0.352	0.364	0.315
	ConvNeXt-B	0.444	0.493	0.462	0.482	0.511	0.452
	SegFormer-B5 + ConvNeXt-B + RF (ours)	**0.646**	**0.666**	**0.638**	**0.693**	**0.678**	**0.669**
	Improvement	**0.256**	**0.280**	**0.226**	**0.326**	**0.206**	**0.289**

Note: VGGNet implementations follow Wang et al. (2021) and Xu et al. (2023), while ResNet implementations are based on Zhang et al. (2018) and Wei et al. (2022). Unlike the binary classification tasks in the referenced studies, our experiments are applied to ten-class classification tasks and used the latest version of VGGNet and ResNet. In the table, models with the best performance are marked in bold. The measure of improvement is the difference between the performance scores of our approach and the VGGNet model.

Table 5
Comparison of different regression models with "SegFormer-B5 + ConvNeXt-B". The highest score are highlighted in bold.

	Regression Method	BeP.	BoP.	DeP.	LiP.	SaP.	WeP.
MAE	LR	0.709	0.618	0.586	0.714	0.874	0.630
	SVR	0.614	0.557	**0.518**	**0.579**	0.668	0.540
	KNR	0.625	0.577	0.529	0.580	0.705	0.533
	AdaBoost	1.279	1.110	1.117	1.148	1.091	0.960
	GBM	0.638	0.576	0.553	0.628	0.665	0.557
	bagging	0.595	0.551	0.523	0.588	0.631	0.538
	MLP	0.886	0.646	0.599	0.701	0.707	0.597
	RF	**0.585**	**0.550**	0.533	0.598	**0.582**	**0.532**
MSE	LR	1.623	1.295	1.069	1.571	1.384	1.335
	SVR	1.571	1.241	**1.041**	1.518	1.217	1.266
	KNR	1.652	1.429	1.188	1.522	1.324	1.343
	AdaBoost	2.399	1.905	1.741	2.283	1.597	1.601
	GBM	1.408	1.291	1.057	1.494	1.051	1.165
	bagging	1.416	1.211	1.049	1.434	**1.050**	1.169
	MLP	1.914	1.241	1.179	1.503	1.109	1.188
	RF	**1.387**	**1.210**	1.076	**1.427**	1.297	**1.163**
R^2	LR	0.585	0.642	0.640	0.662	0.410	0.621
	SVR	0.599	0.657	**0.650**	0.673	0.482	0.640
	KNR	0.578	0.605	0.600	0.672	0.436	0.618
	AdaBoost	0.387	0.473	0.414	0.509	0.320	0.545
	GBM	0.641	0.661	0.644	0.692	0.552	0.669
	bagging	0.639	0.665	0.647	0.692	0.553	0.676
	MLP	0.511	0.657	0.603	0.677	0.528	0.662
	RF	**0.646**	**0.666**	0.638	**0.693**	**0.678**	**0.669**

computed the average, maximum, minimum, and standard deviation for each element, and the deviations from the top-performing SegFormer-B5 (Fig. 5). Notably, substantial variance was observed in elements like buildings, roads, fences, skies, vegetation, and cars. In the Place Pulse 2.0 dataset, variations in the composition of roads, buildings, and sky were nearly or more than 10%, with the FCN model exhibiting a significant 31% discrepancy in building recognition. These disparities could notably impact the accuracy of street view element recognition and subsequently, the evaluation results.

4.4. Perception maps of Tianhe District: diagnosing city quality

For the 36,260 SVIs in Tianhe District dataset, we first employed SegFormer-B5 to obtain the proportions of 19 visual elements in each image and then calculated four objective visual metrics: GVI, SVF, Enclosure and Walkability. Next, we used the "SegFormer-B5 + ConvNeXt-B + RF" approach to predict six subjective perception scores. Fig. 6 provides examples of SVIs with different scores. For each street view point, we averaged the scores from the four directions to derive a consolidated score for that location.

As shown in Table 6, all metrics displayed p-values close to 0, indicating statistically significant global spatial auto-correlation. The six perception metrics exhibit high global Moran's I, suggesting positive spatial dependence and spatial clustering. However, the global Moran's I values for the four objective visual metrics are generally lower, with 'walkability' recording a mere 0.149, suggesting minimal spatial clustering.

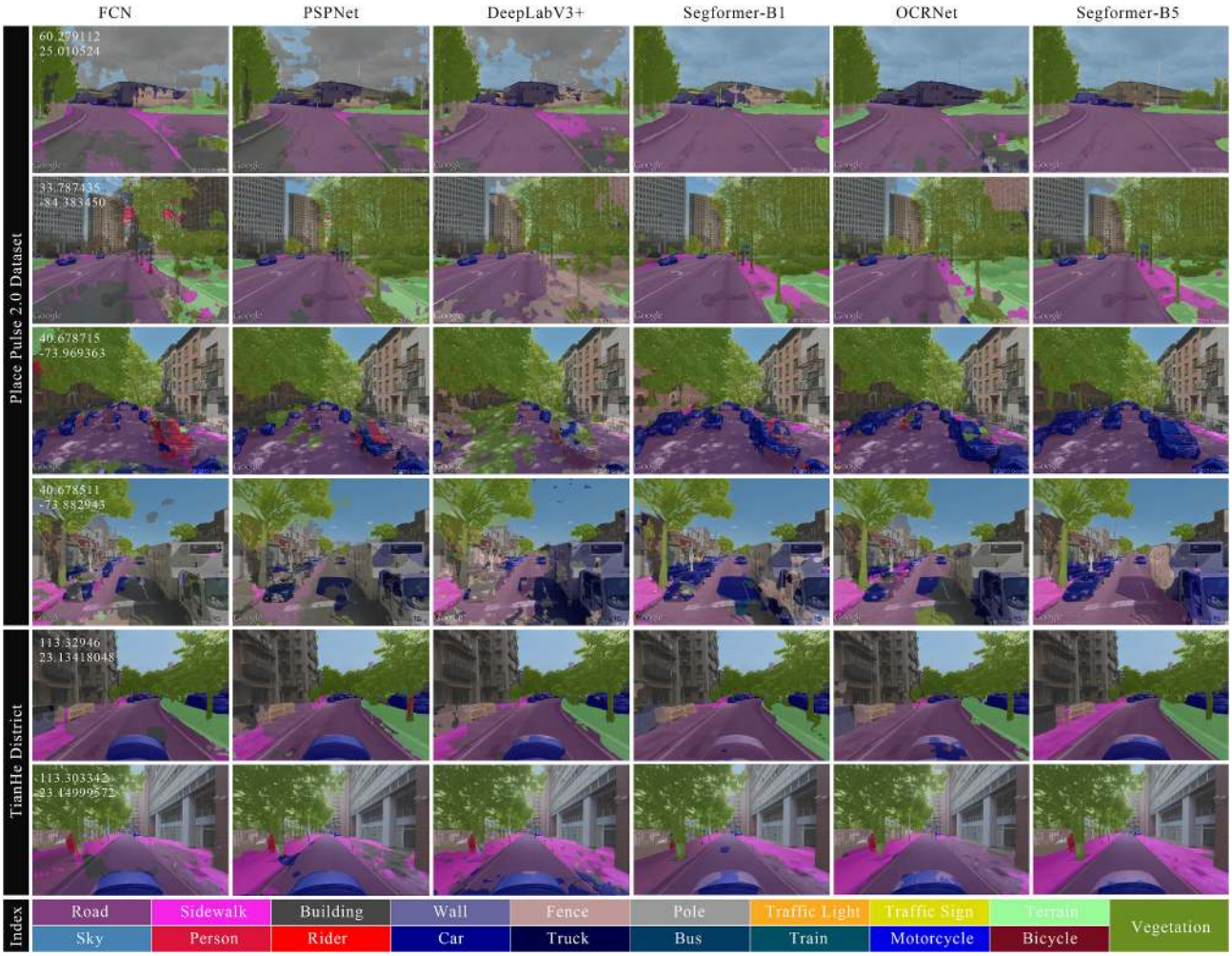


Fig. 3. Semantic segmentation results from different models on SVIs within the Place Pulse 2.0 and Tianhe District datasets.

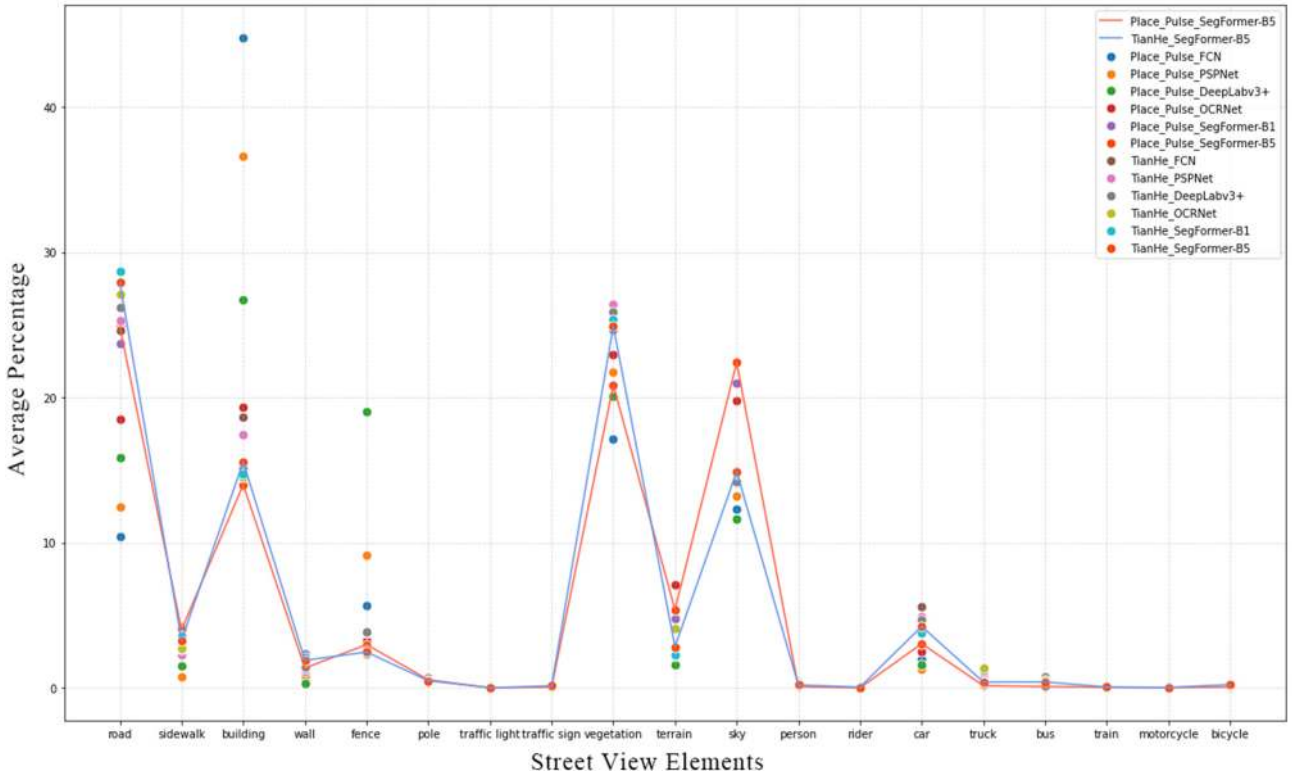


Fig. 4. Average percentage of street view elements identified by different models. Each point represents a different model, and the connecting lines illustrate the trends in changes of street view elements across two datasets.

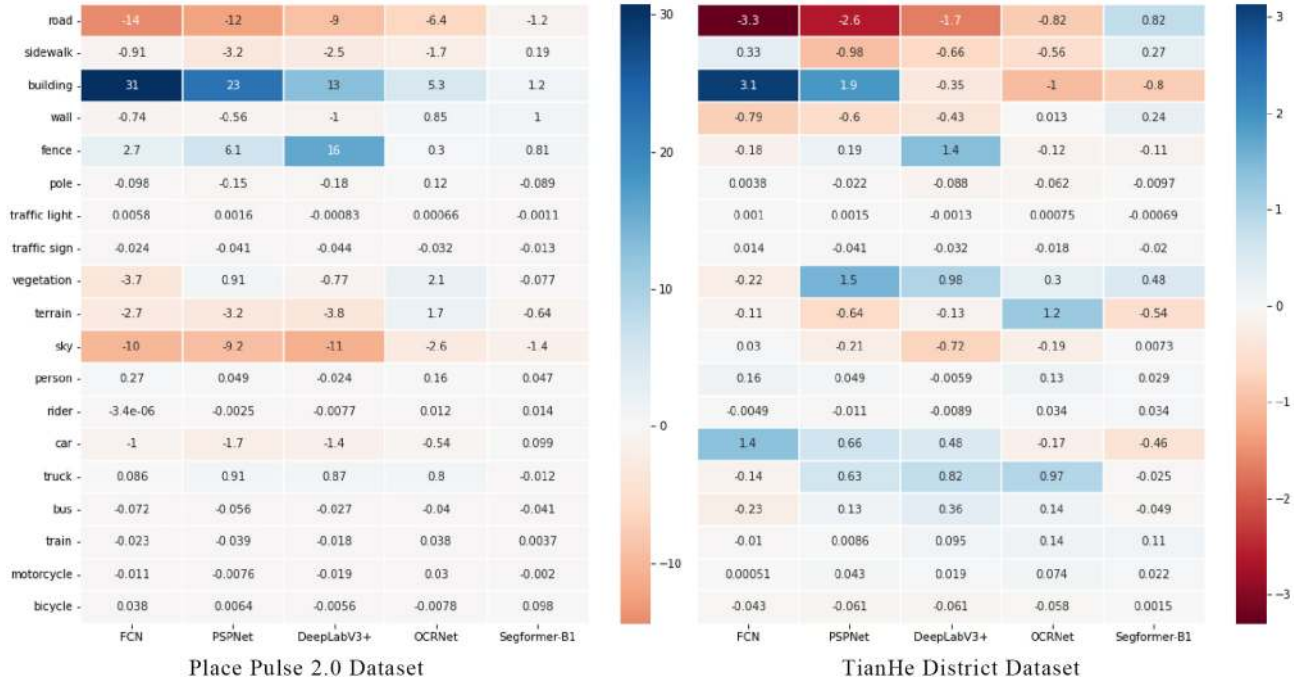


Fig. 5. Deviations from the top-performing SegFormer-B5.

To intuitively present each metric's spatial distribution, we visualized the values at each point using ArcMap 10.6. Through local Moran's I analysis, we classified the points into H–H, L-H, L-L and H-L clusters (Fig. 7). Notably, nine metrics, excluding walkability, displayed distinct and similar distribution patterns in the district. The southwest region exhibited clear spatial clustering with positive perceptions (beautiful, lively, safe, wealthy) and negative perceptions (boring, depressing) showing significant spatial positive correlation, predominantly in H–H and L-L clusters.

Moving from the city center to the suburbs, we observed a decline in positive perceptions, GVI, and enclosure, contrasted with an increase in negative perceptions and SVF. Highways and primary roads manifest



Fig. 6. Selected examples of SVIs from Tianhe District dataset, illustrating varied scoring.

Table 6
Global Moran's I for the ten metrics.

Index	Beautiful	Boring	Depressing	Lively	Safe	Wealthy	GVI	SVF	Enclosure	Walkability
Moran's I	0.557	0.542	0.458	0.625	0.561	0.620	0.400	0.486	0.412	0.149
Z-score	30.807	29.981	25.316	34.519	31.021	34.255	81.666	99.226	84.005	30.572
P value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001

pronounced levels of linear spatial auto-correlation owing to their uniform roadscapes. Compared to the higher environmental complexity of secondary and tertiary roads, these roads exhibit lower GVI and enclosure values, higher SVF values, and generally negative perceptions.

4.5. Exploring the driving objective factors of perception judgement

4.5.1. SHAP visualization results

As determined in Section 4.2 the incorporation of semantic feature vectors enhances the precision of subjective perception prediction, indicating a certain relationship between the two. We further probed into this relationship using SHAP (Fig. 8).

For instance, 'vegetation' significantly influences the 'beautiful' perception, with a higher SHAP value indicating a more substantial positive effect. In essence, amplifying the proportion of vegetation in an image can bolster individuals' preference for the 'beautiful' facet of the environment. Similarly, 'vegetation' positively contributes to diminishing the extent of 'boring' and 'depressing' perceptions while enhancing the 'wealthy' perception. But the importance of 'vegetation' diverges across different perceptions. Similarly, the 'sky' element significantly affects all six perceptions. A greater proportion of sky tends to foster more negative perceptions of the environment. This tendency is associated with the data distribution illustrated in Fig. 7, where images with a higher sky proportion often originate from more distant, less developed, and poorly maintained suburban areas, thereby leading to lower visual quality.

4.5.2. Grad-CAM visualization results

The proportion of a particular element in an image alone cannot determine people's preference for a given scene. Aoki (1991) discovered that the proportion of vegetation exceeding 30% within an image could enhance people's preference, with a critical point around 45%–48%. Therefore, we argue that relying solely on the regression of visual element proportions and perception scores is insufficient in differentiating subjective and objective factors in SVIs. To illustrate this, we analyzed 54 SVIs from Fig. 6 using the Grad-CAM method, revealing ConvNeXt's focus areas in predicting perceptions (Fig. 9).

Similar to human perception, ConvNetXt focused on visual dominant elements in image recognition. These include identifiable objects like fences (1b, 1g, 2b, 5b, 5d) and walls (2a), as well as street furniture such as lighting fixtures (4b), barriers (1b), and sidewalks (2i, 3i). The most detailed or distinctive parts of the image, like building (1c, 3b, 3c) and tunnel facades (2d), and logos on building facades (4a, 5a, 5b) are also emphasized. Specifically, for the 'beautiful' perception, fewer artificial objects shift the model's focus to natural spaces under shrubs and trees (1h, 1i). For 'lively' and 'boring', it concentrates on areas rich in detail, such as building facades (2h, 2i, 5a, 5b, 5h), fences (2d, 5b, 5d) and lighting facilities (5i). 'Safe' perception is influenced by the quality of buildings, with dilapidated buildings (3b, 3c) lowering the score and dense skyscrapers (3g-3i) increasing it. For 'wealthy', high-rise buildings (4f, 4h, 4i), pedestrian walkways (4g), and road markings (4d, 4e, 4f) are crucial. Finally, for 'depressing', the model pays more attention to vegetation, with lower scores correlating with less vegetation (6g-6i)

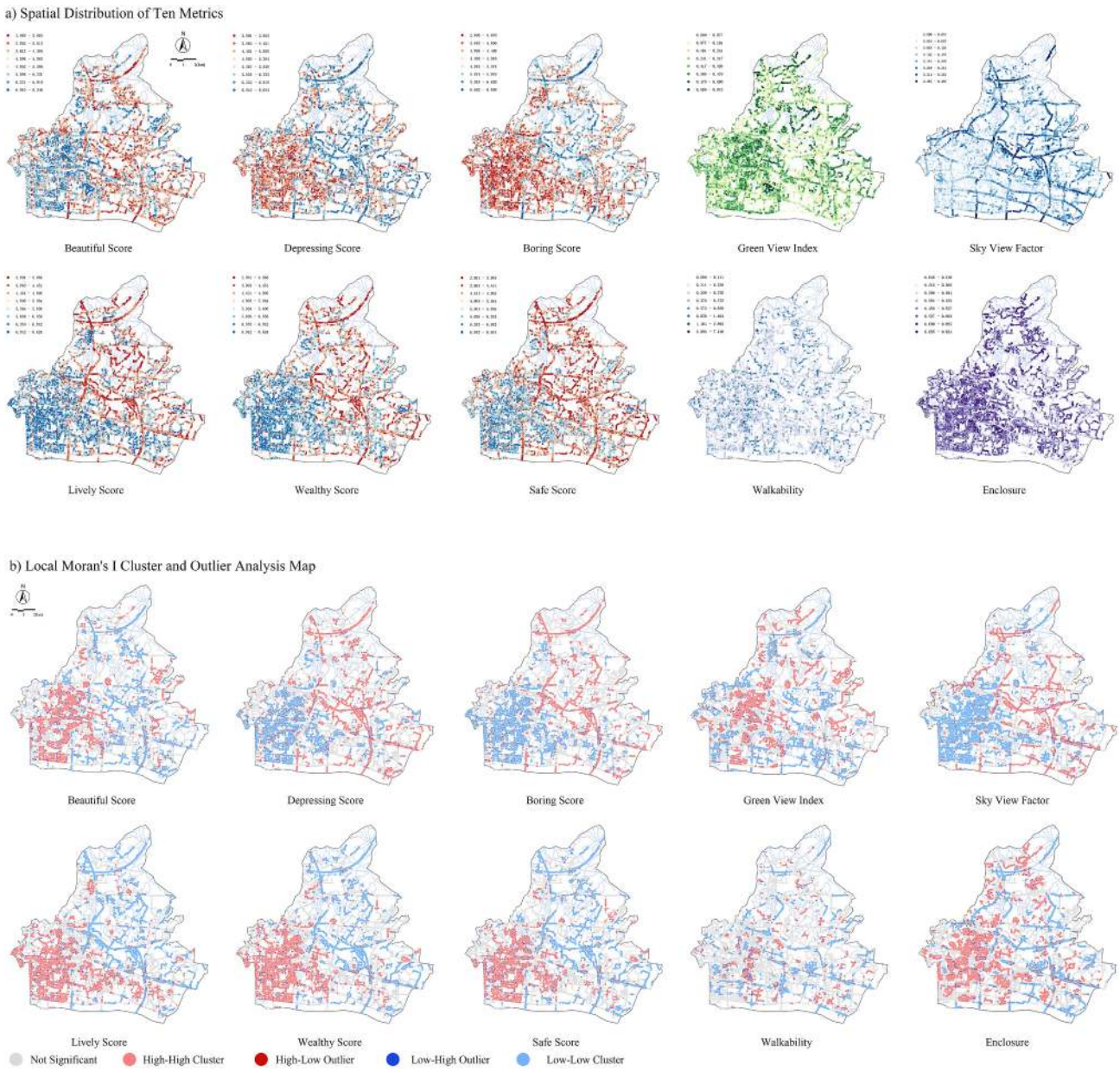


Fig. 7. a) Spatial distribution map of ten metrics in Tianhe district, where red represents lower scores and blue higher scores for six subjective metrics. Deeper colors indicate higher scores for four objective metrics. b) Local Moran's I cluster and outlier analysis map, providing a clear visualization of clustering and outliers for spatial quality identification.

and more hard surfaces and man-made structures.

5. Discussion

5.1. A more accurate approach for assessing the visual qualities of built environments

The "SegFormer-B5 + ConvNeXt-B + RF" approach proposed in our study effectively predicts six subjective perceptions with an average accuracy of 78.56%. This accuracy surpasses that of previous ten-classification study (Ji et al. 2021) and offers more nuanced perception information compared to binary classification approaches (Zhang et al., 2018, Chen et al., 2022, Wei et al., 2022). Overall, this approach can serve as a powerful tool for effectively diagnosing the issues within a city and the generated results can accurately reflect the real quality of the targeted environment.

Additionally, our comparison of six semantic segmentation models revealed that SegFormer-B5 excels in extracting visual element proportions in SVIs, demonstrating high accuracy in both high-resolution and low-resolution images. This finding underscores the importance of incorporating the latest deep learning models in research.

5.2. Interpretation of perception maps to uncover city quality

In perception maps, color variations can visually indicate areas of high quality and areas that are less favored by individuals. This is particularly beneficial for urban designers, as it allows for the rapid pinpointing of regions in need of development or renewal.

In the Tianhe district's southwest, a hub for high-end industries such as economy, finance, technology, and commerce, we observed a clear spatial clustering in the perceptual metrics. Positive perceptions, GVI and enclosure are relatively high, while negative perceptions and SVF tend to be comparatively lower. Moving outward from this core area, there is more undeveloped land, lower urbanization level, higher sky

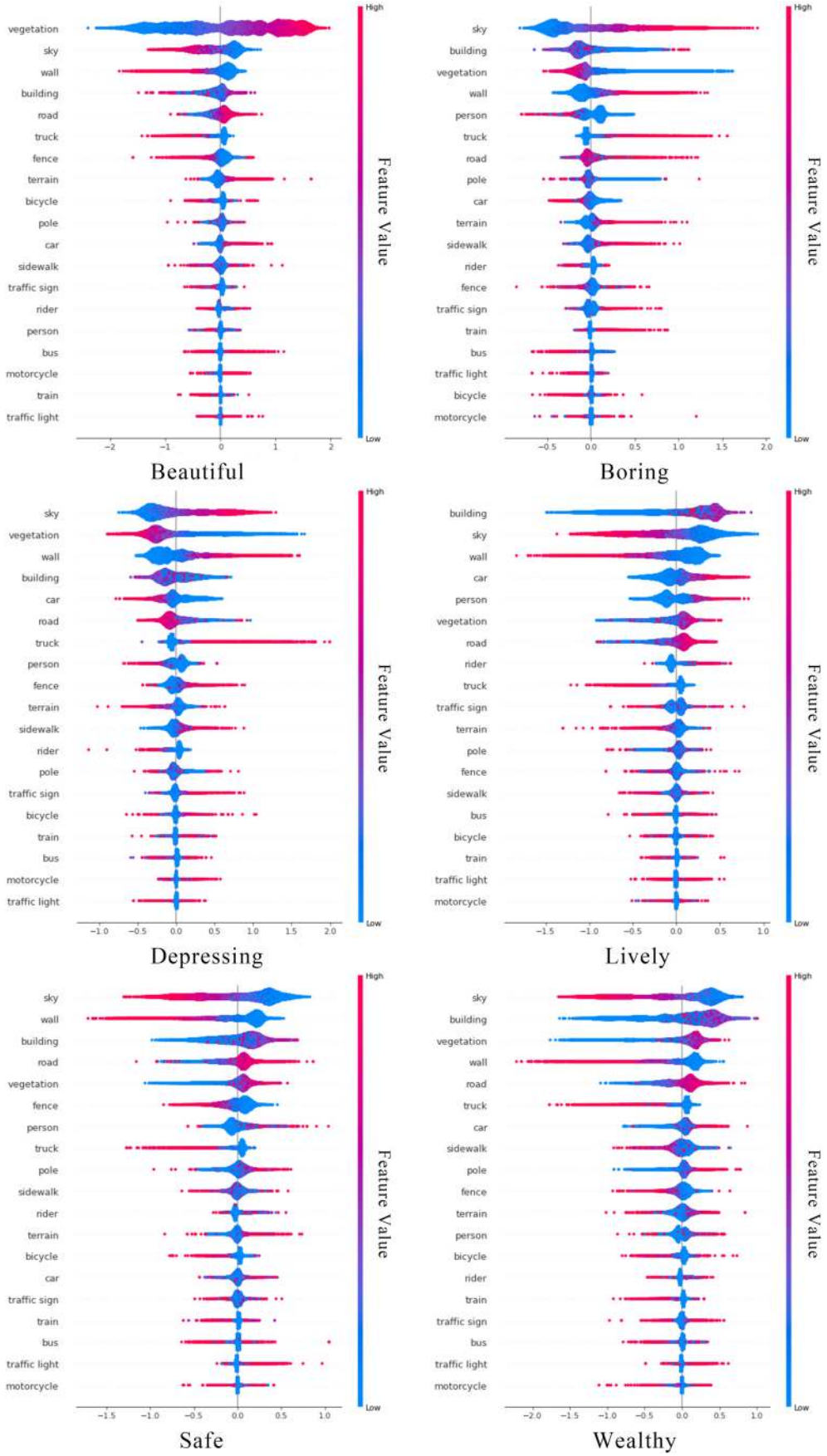


Fig. 8. SHAP visualization depicting feature influence on model predictions: red for positive and blue for negative impacts.

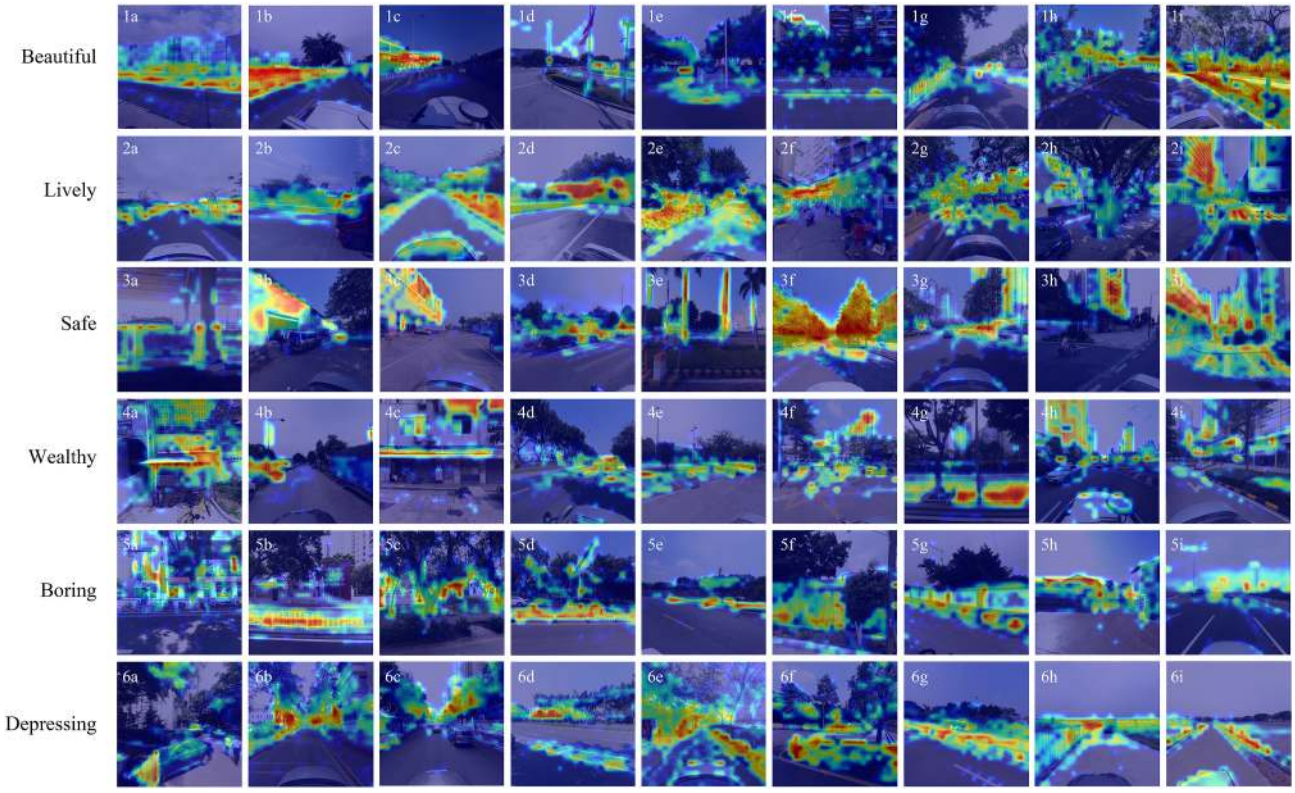


Fig. 9. Grad-CAM visualization highlighting the model's focus areas, which are marked in red.

ratio, and more areas with poor environmental quality. These result in a decrease in positive perceptions, GVI, and enclosure, and an increase in negative perceptions and SVF, corroborating findings from previous studies (Zhang et al., 2018, Wang and Rodiek, 2019, Yao et al., 2019, Wei et al., 2022, Ye et al. 2022).

Interestingly, the six subjective perceptions exhibit similar spatial distribution patterns, indicating consistency in how people perceive positive and negative perceptions. Previous studies have suggested a positive correlation between walkability, urban green spaces (Lu et al. 2018), and street enclosure (Li et al. 2018). However, no obvious clustering pattern of walkability was observed within our study area. This may be due to the prevalence of arcade buildings in Guangzhou, where pedestrian walkways are located underneath. Capturing these walkways when taking pictures from the road can be challenging, resulting in an imprecise assessment of the walkability. Nonetheless, this issue is not attributable to our approach.

5.3. Visualizing the links between visual elements and perception scores for enhanced urban renewal strategies

People's subjective perception is based on their judgment of objective visual elements in the environment, which is related to both the quantity and quality of these elements. Firstly, we found that the increase in the proportion of vegetation positively impacts people's perception, consistent with previous VLA research using photographs, which shows that humans generally prefer lush vegetation (Wang and Rodiek, 2019) and natural landscapes (Sahraoui et al., 2021). However, the impact of vegetation varies in different perceptions, indicating that human perception is the result of the comprehensive consideration of multiple visual elements. Moreover, the increase in the proportion of buildings, for example, can reduce the 'boring' perception and enhance 'lively', 'safe', and 'wealthy' perceptions. This is often observed in bustling city centers (Xia et al., 2020). Walls are negative elements for all six perception dimensions, closely related to GVI and enclosure (Zhu et al., 2022), as observed by Zhang et al. (2018) and Ji et al. (2021). Excessive solid walls or similar obstacles can reduce the visibility and permeability of public spaces. In urban design, minimizing wall height or integrating vegetation with vertical structures can mitigate this effect (Zhu et al., 2022).

Finally, our innovative comparison of the CAM visualization results with eye-tracking technology studies revealed several parallels. For example, Amati et al. (2018) found that people tend to focus on artificial objects, such as lamp posts and garbage cans, which have sharp edges and contrast with the background. Similarly, Li et al. (2020) observed that people's attention is drawn to both attractive and unattractive elements in photos, which is consistent with the attention mechanisms identified in ConvNeXt. In addition, Noland et al. (2017) found that natural features, sidewalks, and street elements are related to image rankings, and in our experiment, vegetation, sidewalks, and urban furniture did receive more attention. At the same time, they found that buildings are neutral, which may depend on whether the building interface is attractive. In our case, low-quality interfaces such as 3b and 3c in Fig. 9 will reduce the perception of safety, while high-quality skyscrapers such as 3g, 3h, 4h, and 4i contribute to the generation of positive perception. Furthermore, Dupont et al. (2014) found that heterogeneous landscapes are quickly scanned because they lack attractive or interesting elements, while heterogeneous landscapes are more diverse and can be explored visually. In our experiment, images with rich details also received more attention (1i, 2i, 3i, 4a, etc.). Finally, ConvNeXt displayed different focus points in different combinations of ground cover, shrubs, and trees, which is similar to the heat maps generated by eye-tracking analysis.

The exploration of objective visual factors influencing subjective perception scores aids in revealing the focal points of human attention when observing landscapes. Upon identifying low-quality areas through perception maps, researchers can further employ methods such as SHAP and CAM to visually pinpoint the specific contributing to the diminished environmental quality. These visualization methods prove highly

advantageous for designers to undertake targeted improvements.

5.4. Advances and limitations

Our approach combines SVIs and deep learning techniques to accurately map the citywide distribution of visual quality metrics in high-density urban contexts. By training on the Place Pulse 2.0 dataset, our model can predict perception scores that reflect the general perception of individuals worldwide. Additionally, the utilization of publicly available street view data from various cities allows our perception prediction model to be rapidly and accurately applied worldwide. Finally, the insights gained from applying SHAP and CAM methods, based on deep learning models, are invaluable. They provide detailed understanding of the underlying causes of urban issues, enabling targeted interventions to improve the urban environments.

This study has several limitations. Firstly, we were unable to study certain SVIs due to their unavailability (Biljecki and Ito, 2021; He and Li, 2021). Future research could expand SVI sources using social media platforms. In addition, while our study developed perceptual prediction models, it didn't focus on datasets that might align more closely with local perceptions, such as those established for Wuhan and Guangzhou in China (Yao et al., 2019, Chen et al., 2022). Future applications of our models could incorporate these local datasets for more nuanced insights. Additionally, combining the subjective and objective metrics in this study with other data sources, such as physiological and psychological health metrics (Wang and Rodiek, 2019), housing prices (Ji et al. 2021), land use types (Wei et al., 2022, Zhu et al., 2022) may provide more exploration space.

6. Conclusions

Streets play a crucial role in the built environment, holding significant value across various dimensions, including transportation, aesthetics, and community interaction (Tang and Long, 2019). This study leverages SVI data and introduces the 'SegFormer-B5 + ConvNeXt-B + RF' deep-learning method to accurately assess visual qualities in large-scale built environments, integrating both objective visual elements and subjective perceptions.

Our approach generates perception maps incorporating six subjective and four objective visual metrics to visually identify areas of high quality and areas that are less favored by individuals. The results show a clear correlation between the spatial distribution of objective visual elements and subjective perceptions, with the city center generally perceived more positively than the suburbs.

In addition, our study offers three new insights into the links between objective visual elements and subjective perceptions. Firstly, we found that the inclusion of semantic feature vectors significantly improves the accuracy of subjective perception prediction models. Secondly, our SHAP analysis offers a detailed ranking and significance of nineteen visual elements in influencing six subjective perceptions. Lastly, we conducted the first-ever comparison between the CAM visualization results and eye-tracking VLA studies, discovering parallels in how deep learning models and humans process visual information. These include paying more attention to artificial objects, both attractive and unattractive elements, and heterogeneous landscapes. These three perspectives are all based on the proposed deep learning approach, which fills the limitations of prior research that only conducted correlation analysis between the proportion of visual elements and perception scores.

Practically, our approach is applicable globally for assessing visual qualities in built environments. It aids in pinpointing low-quality areas and understanding which aspects draw human attention in these areas. This guidance is crucial for formulating evidence-based urban renewal strategies, aiming to create high-quality, sustainable, and inclusive urban living spaces that improve resident well-being.

CRediT authorship contribution statement

Xukai Zhao: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Yuxing Lu:** Writing – review & editing, Methodology. **Guangsi Lin:** Writing – review & editing, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.engappai.2023.107805.

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